

Viet Nguyen

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SUMMARY

My research lies at the intersection of statistics and machine learning, where I seek to understand the behavior of machine learning models from a rigorous theoretical perspective. My recent work focuses on parameter convergence rates in Mixture-of-Experts models. I am also interested in quantum computing, especially quantum optimization and quantum machine learning. My long-term goal is to develop statistical methodologies that enhance the reliability and interpretability of machine learning systems across both classical and quantum computing platforms.

EDUCATION

Hanoi University of Science and Technology (HUST) <i>BSc in Computer Science (Talented Program) — CPA: 3.9/4.0, Top 1% of the cohort</i>	Hanoi, Vietnam 2022 – 2026
High School for Gifted Students, Hanoi University of Science (HUS) <i>High School Diploma (Specialized in Mathematics)</i>	Hanoi, Vietnam 2019 – 2022

EXPERIENCE

Independent Researcher <i>Statistical Machine Learning Group (remote)</i> - Supervisor: Assoc Prof. Nhat Ho, The University of Texas at Austin - Focused on the theoretical analysis of Mixture-of-Experts models - Led two research projects resulting in submissions to NeurIPS 2026, currently under review.	Oct 2025 – Present Hanoi, Vietnam
<i>Trivita AI (remote)</i> - Supervisor: Assoc Prof. Nhat Ho, The University of Texas at Austin - Build AI model for medical image retrieval, focusing on colorectal cancer detection - Fine-tuned DINOv2 with global search and local reranking, achieving the highest mAP among foundation models	Hanoi, Vietnam
Undergraduate Research Assistant <i>Hanoi University of Science and Technology</i> - Supervisor: Dr. Trinh Chien, Hanoi University of Science and Technology - Formulated the communication channel estimation problem using rigorous mathematical analysis - Introduced an evolutionary algorithm to optimize channel quality up to 33% - Led to a paper accepted by IEEE Transactions on Vehicular Technology	Jan 2024 – Sep 2025 Hanoi, Vietnam
<i>Pusan National University (remote)</i> - Supervisor: Dr. Trinh Chien, Hanoi University of Science and Technology - Using quantum computer to solve multi-objective optimization problems - Developed a quantum algorithm that optimized up to $9\times$ faster than state-of-the-art classical methods - Led to a paper accepted by IEEE Transactions on Quantum Engineering	Hanoi, Vietnam

JOURNAL PUBLICATIONS

[Advanced Quantum Annealing for the Bi-Objective Traveling Thief Problem: An \$\varepsilon\$ -Constraint-Based Approach](#)

Viet Nguyen, Nguyen Xuan Tung, Trinh Van Chien, Won-joo Hwang

IEEE Transactions on Quantum Engineering, 2026

[Improved Differential Evolution for Enhancing the Aggregated Channel Estimation of RIS-Aided Cell-Free Massive MIMO](#)

Trinh Van Chien, **Viet Nguyen**, Symeon Chatzinotas, Lajos Hanzo

IEEE Transactions on Vehicular Technology, 2025

[Extending network lifespan and securing effective barrier coverage in 3D underwater wireless sensor networks](#)

Nguyen Thi My Binh, Le Duc Anh Vu, **Nguyen Hoang Viet**, Huynh Thi Thanh Binh, Ho Viet Duc Luong

Cluster Computing, 2026

CONFERENCE PUBLICATIONS

[A Mixture of Experts Vision Transformer for High-Fidelity Surface Code Decoding](#)

Viet Nguyen, Manh Hung Nguyen, Hoang Ta, Van Khu Vu, Yeow Meng Chee

IEEE International Symposium on Information Theory (ISIT) 2026

PREPRINTS

[Rethinking Multinomial Logistic Mixture of Experts with Sigmoid Gating Function](#)

Tuan Minh Pham*, Thinh Cao*, **Viet Nguyen***, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**

Under Review at NeurIPS 2026

[A Statistical Theory of Gated Attention through the Lens of Hierarchical Mixture of Experts](#)

Viet Nguyen*, Tuan Minh Pham*, Thinh Cao*, Huy Nguyen, Nhat Ho**, Alessandro Rinaldo**

Under Review at NeurIPS 2026

ACHIEVEMENT

Academic Encouragement Scholarship (5 semesters)

2022 – 2026

Scholarships for students with excellent academic achievements at HUST

Best Presentation Award

Sep 2025

SoICT Summer School on AGENTIC AI

- *Project name: First steps toward Evaluating RAG through Agent-based Attack and Defense*

Vietnam Math Competition for University student

2025

The most prestigious math competition for university students in Vietnam (First Prize – ranked 8th)

Vietnam Math Competition for University student

2023

The most prestigious math competition for university students in Vietnam (First Prize – ranked 4th)

Vietnam Mathematical Olympiad

2020

The most prestigious math competition for high school students in Vietnam (Second Prize – ranked 13th)

PROFESSIONAL SERVICES

Reviewer at

- Transactions on Machine Learning Research ([TMLR](#))

LANGUAGE

Vietnam: Native

English: Full professional proficiency, IELTS 7.0

REFERENCES

[Prof. Nhat Ho](#)

Associate Professor, The University of Texas at Austin

[Trinh Chien](#)

Assistant Professor, Hanoi University of Science and Technology

[Hoang Ta](#)

Assistant Professor, Hanoi University of Science and Technology